

Client Proposal

Data-Driven Publishing Strategy for Steam Game Releases

Entity

Pixel Publishing is a mid-sized independent video game publisher that releases approximately **3-6** PC games on Steam each year. Developing and marketing new titles requires significant investment, so leadership must carefully decide which projects to fund and how to position them before launch. Rather than relying solely on intuition or industry trends, the company wants to use historical Steam data to guide future publishing decisions.

Problem Definition

Because the company can only publish a handful of titles each year, selecting the wrong project represents a significant financial risk. Management wants to know whether focusing on historically successful genres is sufficient, or whether combining multiple pre-launch characteristics produces more reliable publishing decisions.

The analysis should address two questions:

1. Genre-based strategy

After accounting for price, release period, platform availability, and other observed game characteristics, are particular genres consistently associated with a higher likelihood of commercial success?

2. Predictive approach

Can a model using information available before or around launch classify a planned title as likely to achieve a high ownership tier with sufficient reliability to support an initial go/no-go decision? The model may consider features such as price, genres, platforms, language support, developer, publisher, and other metadata determined to be temporally appropriate.

The two approaches should be compared to determine whether a multivariable predictive model provides meaningful decision value beyond a simpler genre-based strategy.

Stakeholder / Audience

The primary stakeholder is the **Director of Publishing** at Pixel Publishing. The results will support decisions about which projects to approve, how they should be priced and positioned, and where development and marketing resources should be invested before release.

Datasets

Steam Games Dataset:

<https://www.kaggle.com/datasets/fronkongames/steam-games-dataset/data>

This dataset consists of 125,855 games published on Steam with information including release date, price, supported platforms, supported languages, developers, publishers, genres, categories, tags, review statistics, estimated ownership, and player engagement metrics.

Dataset Limitations

The Steam Games Dataset is a current snapshot rather than a historical record of each game's launch conditions. Estimated ownership is reported as a range rather than an exact figure, and most games occupy the lowest ownership bucket. Fields such as current price, tags, categories, and DLC count may also have changed after launch and must be evaluated for potential temporal leakage.

Metacritic score and Score rank are populated for only a small fraction of titles and will not be central to the analysis. Commercial success will therefore be represented through ownership-tier classification, with review volume used only as a secondary outcome check rather than as a pre-launch model input.

Scope Filtering

Where possible, non-game listings such as demos, utilities, and soundtracks will be identified and excluded based on available metadata. Extremely large AAA-scale titles may be treated separately or excluded using a documented criterion determined during exploratory data analysis.

Post-launch outcomes such as review count, estimated ownership, peak concurrent users, and playtime will not be used to determine eligibility for the core sample, because filtering on these outcomes could bias the analysis.

Success Criteria

The project will be considered successful if it:

- Determines whether genre remains meaningfully associated with ownership-tier success after accounting for other observed characteristics.
- Evaluates whether a multivariable model provides useful predictive performance beyond a simpler genre-based baseline.
- Defines and justifies an ownership-based success threshold based on the observed ownership distribution.
- Identifies which attributes available before or at launch are most strongly associated with predicted success (ownership-tier classification) without interpreting observational relationships as causal effects.
- Produces recommendations that the Director of Publishing can use when screening and positioning future game concepts.

TI;DR

Pixel Publishing wants to determine whether selecting games based primarily on genre is sufficient, or whether combining multiple pre-launch game characteristics can produce more reliable publishing decisions. Using historical Steam data, the project will compare a genre-only strategy against a multivariable predictive model to estimate a game's likelihood of achieving a high ownership tier.